

Project Waves and Instabilities: MHD waves in a magnetic flux tube

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A magnetic flux tube is a bundle of magnetic field lines with constant magnetic flux. Such a structure is ubiquitous in the solar atmosphere (such as coronal loops, sunspots, spicules, etc.), serving as the wave-guides of various MHD waves. In this project, we will use the ideal magnetohydrodynamics (MHD) model to study linear MHD waves in a magnetic flux tube using a cylindrical coordinate system (r, θ, z) (as shown in Figure 2). The equilibrium magnetic field, density, and pressure are both uniform inside and outside the loop, but have different values.

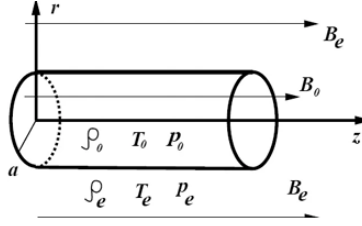


Figure 1: A magnetic flux tube of radius a embedded in a magnetised plasma. Figure taken from Nakariakov Verwichte (2005).

1 Question 1

Using the boundary conditions for displacement and pressure to derive the following dispersion relation:

$$\frac{M_0 a}{\rho_0 (k_z^2 c_{A0}^2 - \omega^2)} \frac{I'_m(M_0 a)}{I_m(M_0 a)} = \frac{M_e a}{\rho_e (k_z^2 c_{Ae}^2 - \omega^2)} \frac{K'_m(M_e a)}{K_m(M_e a)} \quad (1.1)$$

Where $M_0^2 = \frac{(k_z^2 c_{A0}^2 - \omega^2)(k_z^2 c_{s0}^2 - \omega^2)}{(c_{s0}^2 + c_{A0}^2)(k_z^2 c_{t0}^2 - \omega^2)}$, $M_e^2 = \frac{(k_z^2 c_{Ae}^2 - \omega^2)(k_z^2 c_{se}^2 - \omega^2)}{(c_{se}^2 + c_{Ae}^2)(k_z^2 c_{te}^2 - \omega^2)}$. c_s and c_A stands for sound speed and Alfvén speed, and the tube speed c_t satisfies $\frac{1}{c_t^2} = \frac{1}{c_s^2} + \frac{1}{c_A^2}$.

We start with an explanation of the system:

- $r < a$: we are **inside the cylinder**, we consider the following measures as uniform B_0, T_0, ρ_0, p_0 .
- $r > a$: we are **outside the cylinder**, we consider the following measures as uniform B_e, T_e, ρ_e, p_e .

Where a is the cylinder's radius noting that we consider the cylindrical coordinate system (r, θ, z) with $f(r, \theta, z, t) = \hat{f}(r)e^{i(k_z z + m\theta - \omega t)}$.

Since we have two different regions we have the following behaviors:

- $r < a$: inside the tube.

We have to consider the **Modified Bessel of Type I**

$$p_{T_0}(r) = A_0 I_m(M_0 r) \quad (1.2)$$

where A_0 is a constant and $M_0^2 = \frac{(k_z^2 c_{A_0}^2 - \omega^2)(k_z^2 c_{s_0}^2 - \omega^2)}{(c_{s_0}^2 + c_{A_0}^2)(k_z^2 c_{t_0}^2 - \omega^2)}$.

- $r > a$: outside the tube.

The energy of the waves has to decrease to infinity ($r \rightarrow \infty$). We have to consider the **Modified Bessel of Type II**

$$p_{T_e}(r) = A_e K_m(M_e r) \quad (1.3)$$

where A_e is a constant and $M_e^2 = \frac{(k_z^2 c_{A_e}^2 - \omega^2)(k_z^2 c_{s_e}^2 - \omega^2)}{(c_{s_e}^2 + c_{A_e}^2)(k_z^2 c_{t_e}^2 - \omega^2)}$.

Then since radial displacement ξ is proportional to $\frac{\partial p_T(r)}{\partial r}$ (the perturbed pressure derivative), in our specific case we get:

$$\xi_r(r) = \frac{1}{\rho(k_z^2 c_A^2 - \omega^2)} \frac{\partial p_T(r)}{\partial r}$$

We have to consider the formula in the two different regions:

- $r < a$:

$$\xi_r(r) = \frac{1}{\rho_0(k_z^2 C_{A_0}^2 - \omega^2)} \frac{\partial p_{T_0}(r)}{\partial r} \quad (1.4)$$

- $r > a$:

$$\xi_r(r) = \frac{1}{\rho_e(k_z^2 C_{A_e}^2 - \omega^2)} \frac{\partial p_{T_e}(r)}{\partial r} \quad (1.5)$$

Next, we evaluate the derivatives of $\frac{\partial p_{T_0}(r)}{\partial r}$ and $\frac{\partial p_{T_e}(r)}{\partial r}$:

$$\begin{aligned} \frac{\partial p_{T_0}(r)}{\partial r} &= \frac{\partial(A_0 I_m(M_0 r))}{\partial r} = A_0 I'_m(M_0 r) M_0 \\ \frac{\partial p_{T_e}(r)}{\partial r} &= \frac{\partial(A_e K_m(M_e r))}{\partial r} = A_e K'_m(M_e r) M_e \end{aligned}$$

Getting at the end:

- $r < a$:

$$\xi_{r_0}(r) = \frac{1}{\rho_0(k_z^2 c_{A0}^2 - \omega^2)} A_0 M_0 I'_m(M_0 r) \quad (1.6)$$

- $r > a$:

$$\xi_{r_e}(r) = \frac{1}{\rho_e(k_z^2 c_{Ae}^2 - \omega^2)} A_e M_e K'_m(M_e r) \quad (1.7)$$

What physical considerations dictate the selection of I_m for the internal region and K_m for the external plasma?

The choice of the modified Bessel functions I_m and K_m is dictated by the physical boundary conditions: we require the pressure perturbation to be finite at the origin ($r \rightarrow 0$) and the wave energy to vanish at infinity ($r \rightarrow \infty$). We start with the solution of the modified Bessel:

$$P_T(r) = C_1 I_m(Mr) + C_2 K_m(Mr)$$

However, pressure and position can't be infinite so:

- $I_m(x)$: for $x \rightarrow 0$ goes to zero, for $x \rightarrow \infty$ is infinite
- $K_m(x)$: for $x \rightarrow 0$ is infinite, for $x \rightarrow \infty$ goes to zero

So, at the end we get:

- for $r < a$: $C_2 = 0$ and $p_{T_0}(r) \propto I_m(M_0 r)$
- for $r > a$: $C_1 = 0$ and $p_{T_e}(r) \propto K_m(M_e r)$

We can now consider the **boundary conditions**:

$$[\xi_r] = 0 \quad [p_T] = 0 \quad \text{at } r = a$$

From these conditions we are able to get the following constraints:

- **Continuity of total pressure:** $p_{T_0}(a) = p_{T_e}(a)$

$$A_0 I_m(M_0 a) = A_e K_m(M_e a) \quad (1.8)$$

- **Continuity of the real displacement:** $\xi_{r_0}(a) = \xi_{r_e}(a)$

$$\frac{A_0 M_0}{\rho_0(k_z^2 c_{A0}^2 - \omega^2)} I'_m(M_0 a) = \frac{A_e M_e}{\rho_e(k_z^2 c_{Ae}^2 - \omega^2)} K'_m(M_e a) \quad (1.9)$$

We have decided to impose the continuity of the normal displacement ($[\xi_r] = 0$) to ensure that the interface between the tube and the external plasma does not tear apart. Furthermore, we try to satisfy the continuity of the total pressure perturbation ($[p_T] = 0$) to maintain force balance across the boundary of the

flux tube. To be able to do the calculation we divide Eq.(1.9) by Eq.(1.8):

$$\frac{M_0 \cancel{A_0}}{\rho_0(k_z^2 c_{A0}^2 - \omega^2)} \frac{I'_m(M_0 a)}{I_m(M_0 a)} \frac{1}{\cancel{A_0}} = \frac{M_e \cancel{A_e}}{\rho_e(k_z^2 c_{Ae}^2 - \omega^2)} \frac{K'_m(M_e a)}{K_m(M_e a)} \frac{1}{\cancel{A_e}}$$

$$\frac{M_0}{\rho_0(k_z^2 c_{A0}^2 - \omega^2)} \frac{I'_m(M_0 a)}{I_m(M_0 a)} = \frac{M_e}{\rho_e(k_z^2 c_{Ae}^2 - \omega^2)} \frac{K'_m(M_e a)}{K_m(M_e a)}$$

Then at the end we multiply both the right-hand and left-hand side for a :

$$\boxed{\frac{M_0 a}{\rho_0(k_z^2 c_{A0}^2 - \omega^2)} \frac{I'_m(M_0 a)}{I_m(M_0 a)} = \frac{M_e a}{\rho_e(k_z^2 c_{Ae}^2 - \omega^2)} \frac{K'_m(M_e a)}{K_m(M_e a)}} \quad (1.10)$$

2 Question 2

Solve the dispersion relation numerically and plot the phase speed as a function of the wave number k_z for the sausage and kink mode ($m = 0, 1$). Visualize the eigenfunction for these two modes for a specific value $k_z a = 1$ and an appropriate density constant.

Hint: For the calculation you can use the paper of Edwin, P. M. and Roberts, B. Wave Propagation in a Magnetic Cylinder, 88, 179 (1983) or Chapter 6 in the book MHD waves in the solar atmosphere by Bernard Roberts.

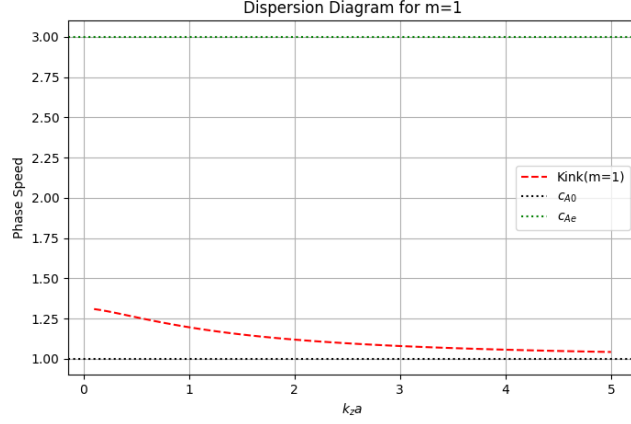
2.1 Methodology and Code Description

To solve the dispersion relation derived in Question 1, a numerical root-finding algorithm was implemented in Python (*scipy.optimize.brentq* function). We then assigned specific values to model the plasma parameters, considering an internal density $\rho_0 = 1$, an external density $\rho_e = 0.1$, an internal Alfvén speed $c_{A0} = 1$ and an external Alfvén speed $c_{Ae} = 3$ (with sound speeds $c_{s0} = 0.5$ and $c_{se} = 0.5$). While evaluating the parameter M_0^2 , we studied the physical transition between different wave regimes:

- **Surface Waves** ($M_0^2 > 0$): the total pressure perturbation is described by type I modified Bessel functions because the phase speed falls outside the body wave regime.
- **Body Waves** ($M_0^2 < 0$): we use the type I ordinary Bessel functions, indeed since v_{ph} falls within the range (c_{A0}, c_{Ae}) , M_0^2 becomes negative resulting in an oscillatory radial structure inside the tube.

2.2 Dispersion Curve: Kink Mode

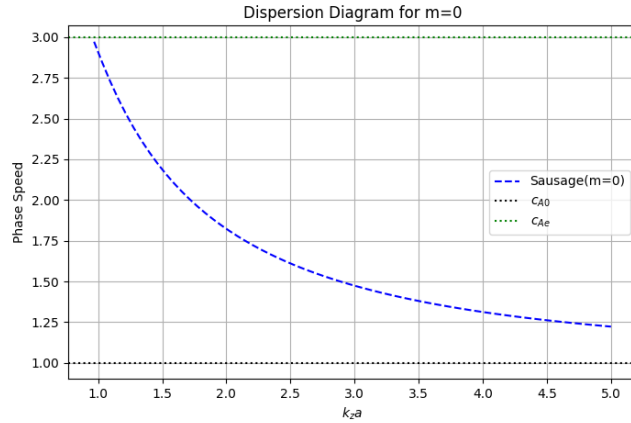
We illustrate the dispersion relation for **trapped MHD waves**, plotting the dimensionless phase speed against the dimensionless wavenumber $k_z a$. For the kink mode ($m = 1$) a trapped solution exists for almost all the values of $k_z a$. As shown in the dispersion diagram, when $k_z a \rightarrow 0$ the phase speed approaches

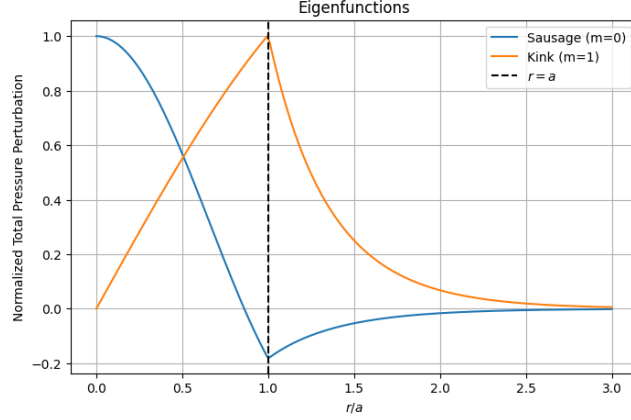


the theoretical kink speed c_k . As the wavenumber increases, the phase speed asymptotically decreases getting closer to the internal Alfvén speed c_{A0} . The trapped nature of this mode is confirmed by the fact that the phase speed is strictly **bounded** between the internal and external Alfvén speed.

2.3 Dispersion Curve: Sausage Mode

The behavior of the sausage mode ($m = 0$) differs from the kink mode because of a low-wavenumber cutoff. As illustrated in the dispersion diagram, the sausage mode can't exist for small values of $k_z a$. In our case the cutoff occurs at approximately $k_z a = 0.9$. Below this critical wavenumber, the wave energy is not confined in within the tube and radiates into the external plasma, going from a trapped mode to a leaky one. For $k_z a > cutoff$ the phase speed decreases, getting closer to the internal Alfvén speed $c_{A0} = 1$ asymptotically as $k_z a$ keeps increasing.





2.4 Eigenfunction of the Sausage Mode

We decided to study the radial profile of the normalized total pressure perturbation for both modes at a specific dimensionless wavenumber $k_z a = 1$. In this case, both modes have a phase speed v_{ph} within the range getting $M_0^2 < 0$ and $M_e^2 > 0$. Both eigenfunctions show a fast body wave structure. Indeed for $r < a$, the plot has a behavior described by ordinary Bessel functions. Then at the boundary ($r = a$) it is continuous. At the end for $r > a$, the pressures of both the modes go to zero in an exponential way. This confirms that the sausage mode is a fully formed trapped mode at this wavenumber, consistent with the cutoff phenomenon.

3 Question 3

Observations of the solar atmosphere often suggest that magnetic flux tubes may be twisted, possessing a magnetic field component B_θ in the azimuthal direction $\vec{e}_\theta$. The model for a twisted magnetic flux tube is shown in Figure 2. To satisfy the pressure balance, we have:

$$\frac{d}{dr}\left(p_0 + \frac{B^2}{2\mu_0}\right) = -\frac{B_\theta^2}{\mu_0 r} \quad (3.1)$$

Here we consider the special case of an equilibrium with uniform twist $\vec{B}(r) = A_r \vec{e}_\theta + B_0 \vec{e}_z$ for $r < a$. Try to derive the dispersion relation of MHD waves in such a case. The relation should reduce to Eq1.

Hint: If you find it too difficult, you can check the paper Bennett, Roberts, and Narain (1999).

In this case the magnetic tube flux may be twisted having B_θ in $\vec{e}_\theta$ direction, so at this point we have to satisfy the pressure balance.

We start again considering the two different regions:

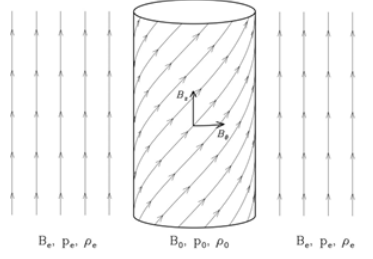


Figure 2: The equilibrium configuration for a twisted tube embedded within an untwisted magnetized plasma

- $r < a$: in Question 1 we only had $B = (0; 0; B_0)$, but in this case we also have a component along $\vec{e}_\theta$ so since $\vec{B}(r) = 0\vec{e}_r + A_r\vec{e}_\theta + B_0\vec{e}_z$ we get $B = (0; B_\theta; B_0) \rightarrow (0; Ar; B_0)$
- $r > a$: since we don't have anything in $\vec{e}_\theta$ direction we consider $B = (0; 0; B_e)$

We conclude that p_0 in the tube ($r < a$) can't be constant so now the derivation is different from zero:

$$\begin{aligned} \frac{d}{dr} \left(p_0 + \frac{B^2}{2\mu_0} \right) &= -\frac{B_\theta^2}{\mu_0 r} \\ \frac{d}{dr} \left(p_0 + \frac{A^2 r^2 + B_0^2}{2\mu_0} \right) &= -\frac{A^2 r^2}{\mu_0 r} \end{aligned}$$

Noting that $B_\theta = A^2 r^2$ and $B^2 = A^2 r^2 + B_0^2$. Then we continue with the derivation:

$$\begin{aligned} \frac{dp_0}{dr} + \frac{A^2 r}{\mu_0} + \frac{d}{dr} \frac{B_0^2}{2\mu} &= -\frac{A^2 r^2}{\mu_0 r} \\ \frac{dp_0}{dr} + \frac{A^2 r}{\mu_0} &= -\frac{A^2 r^2}{\mu_0 r} = -\frac{A^2 r}{\mu_0} \\ \frac{dp_0}{dr} &= -\frac{2A^2 r}{\mu_0} \end{aligned}$$

Getting at the end for $r \leq a$:

$$p_0(r) = p_0(0) - \frac{A^2 r^2}{\mu_0}$$

where $p_0(0)$ is the plasma pressure at the centre of the twisted tube.

Noting that $\frac{dp_0}{dr} < 0$, the pressure decreases as we move away from the center. The force in Alfrén and magnetic acoustic waves depends on how plasma twist the magnetic field lines. In order to study this we can use $B \cdot \nabla$.

$$B \cdot \nabla = \frac{\partial B_r}{\partial r} + \frac{\partial}{\partial \theta} \frac{B_\theta}{r} + \frac{\partial B_z}{\partial z} = \frac{0\partial}{\partial r} + \frac{\partial A r}{\partial \theta} \frac{1}{r} + \frac{\partial B_0}{\partial z} = A \frac{\partial}{\partial \theta} + B_0 \frac{\partial}{\partial z} \quad (3.2)$$

How can we approximate the derivation?

For these kind of waves we can assume that $\frac{\partial}{\partial \theta} \rightarrow im$ and $\frac{\partial}{\partial z} \rightarrow ik_z$. So, the following equivalence is satisfied:

$$imA + ik_z B_0 = i(Am + B_0 k_z)$$

This happens because we are considering an uniform twist so now it is not depending on r anymore. We can notice how the coefficients are constant, so we can use the Bessel equation.

We consider again the Bessel equation:

- $r < a$: we start by requiring that p_T is finite at $r = 0$ giving the following result:

$$p_T = \alpha_0 I_m(m_0 r)$$

where α_0 is a constant. In general m_0 may be real, purely imaginary or complex. As a consequence for $m_0^2 < 0$ it is more convenient to write $n_0^2 = -m_0^2 > 0$ then the solution becomes:

$$p_T(r) = \alpha_0 J_m(n_0 r) \quad (3.3)$$

- $r > a$: doesn't have the twist ($B_\theta = 0$), so the expression for m_0^2 can be reduced to $m_0^2 = k_z^2$. Then requiring that as $r \rightarrow \infty$, p_T is bounded, giving:

$$p_T(r) = \alpha_e K_m(|k_z| r) \quad (3.4)$$

Then we can apply the new boundary conditions, first we must ensure that the normal component of the perturbation ξ_r remains continuous across the unperturbed tube boundary $r = a$, in addition also the total Lagrangian pressure is conserved across the perturbed boundary. This leads to the two following boundary conditions:

$$\begin{aligned} [\xi_r]^{inner} &= [\xi_r]^{outer} \\ [p_T - \frac{B_\theta^2}{\mu_0 a} \xi_r]^{inner} &= [p_T - \frac{B_\theta^2}{\mu_0 a} \xi_r]^{outer} \end{aligned}$$

At this point it is quite important to notice that the Alfvén frequency ω_A and coefficients D , rC_1 and C_3 become constants getting:

$$\begin{aligned} \omega_A &= \frac{1}{(\mu\rho)^{1/2}}(mA + k_z B_0) \\ D &= \rho(\omega^2 - \omega_A^2) \\ C_1 &= -\frac{2mA}{\mu r}(mA + k_z B_0) \\ C_3 &= D^2 - \frac{4A^2}{\mu}\rho\omega_A^2 \end{aligned}$$

Then for a uniformly twisted tube, we have the following equation:

$$\frac{d^2 p_T}{dr^2} + \frac{1}{r} \frac{dp_T}{dr} - \left(\frac{m^2}{r^2} + m_0^2 \right) p_T = 0 \quad (3.5)$$

where $m_0^2 = k_z^2 \left(1 - \frac{4A^2 \omega^2}{\mu \rho (\omega^2 - \omega_A^2)^2} \right)$. Introducing the new values we start to study the value of ξ_r in the region:

- $r > a$:

$$\xi_r(a) = \frac{1}{\rho_e (\omega^2 - \omega_A^2)} \alpha_e |k_z| K'_m(|k_z|a)$$

- $r < a$: finding ξ_r is more difficult in this case because inside the flux the magnetic field is different, so we can follow these steps:

$$\begin{aligned} \xi_r(r) &= \frac{1}{C_3} \left(D \frac{dp_T}{dr} + C_1 p_T \right) \\ \xi_r(r) &= \frac{1}{C_3} (D \alpha_0 m_0 I'_m(m_0 r) + C_1 \alpha_0 I_m(m_0 r)) \\ \xi_r(a) &= \alpha_0 \left[\frac{D m_0 I'_m(m_0 a) + C_1 I_m(m_0 a)}{C_3} \right] \end{aligned}$$

Now we can consider again the two boundary conditions:

$$\begin{aligned} \alpha_0 \left[\frac{D_0 m_0 I'_m(m_0 a) + C_1(a) I_m(m_0 a)}{C_3} \right] &= \alpha_e \left[\frac{|k_z| K'_m(|k_z|a)}{D_e} \right] \\ \alpha_0 I_m(m_0 a) - \frac{A^2 a}{\mu_0} \left(\alpha_e \frac{|k_z| K'_m(|k_z|a)}{D_e} \right) &= \alpha_e K_m(|k_z|a) \end{aligned}$$

By grouping all the terms with α_e in the right hand side we get:

$$\alpha_0 I_m(m_0 a) = \alpha_e \left[K_m(|k_z|a) + \frac{A^2 a}{\mu_0 D_e} |k_z| K'_m(|k_z|a) \right] \quad (3.6)$$

We then divide the first equation by the second one getting:

$$\frac{\frac{D_0}{C_3} m_0 I'_m + \frac{C_1}{C_3} I_m}{I_m} = \frac{\frac{|k_z| K'_m}{D_e}}{K_m + \frac{A^2 a}{\mu_0 D_e} |k_z| K'_m} \quad (3.7)$$

We then multiply both the right-hand side and the left-hand side for a .

We now consider the right hand side getting:

$$RHS = \frac{\frac{|k_z| a K'_m}{K_m}}{D_e + \frac{A^2}{\mu_0} \frac{|k_z| a K'_m}{K_m}} = \frac{\frac{|k_z| a K'_m}{K_m}}{\rho_e (\omega^2 - \omega_{Ae}^2) + \frac{A^2}{\mu_0} \frac{|k_z| a K'_m}{K_m}}$$

On the other hand if we do the same for the left-hand side we get:

$$\begin{aligned}
LHS &= \frac{D_0 \frac{m_0 a I'_m}{I_m} + C_1 a}{C_3} = \\
&= \frac{\rho_0 (\omega^2 - \omega_{A0}^2) \frac{m_0 a I'_m}{I_m} - \frac{2m\rho_0 A \omega_{A0}}{\sqrt{\mu_0 \rho_0}}}{\rho_0^2 (\omega^2 - \omega_{A0}^2)^2 - 4 \frac{A^2}{\mu_0} \rho_0 \omega_{A0}^2} = \\
&= \frac{(\omega^2 - \omega_{A0}^2) \frac{m_0 a I'_m}{I_m} - 2m\omega_{A0} \frac{A}{\sqrt{\mu_0 \rho_0}}}{\rho_0 [(\omega^2 - \omega_{A0}^2)^2 - 4\omega_{A0}^2 \frac{A^2}{\mu_0 \rho_0}]}
\end{aligned}$$

At this point we impose $LHS = RHS$, getting as a final result:

$$\frac{(\omega^2 - \omega_{A0}^2) \frac{m_0 a I'_m}{I_m} - 2m\omega_{A0} \frac{A}{\sqrt{\mu_0 \rho_0}}}{(\omega^2 - \omega_{A0}^2)^2 - 4\omega_{A0}^2 \frac{A^2}{\mu_0 \rho_0}} = \frac{\frac{|k_z| a K'_m}{K_m}}{\frac{\rho_e}{\rho_0} (\omega^2 - \omega_{Ae}^2) + \frac{A^2}{\mu_0 \rho_0} \frac{|k_z| a K'_m}{K_m}} \quad (3.8)$$

We conclude by showing that this can be reduced to Eq. (1.1). We start by assuming that $A = 0$ which implies $m_0 = |k_z|$, simplifying our equation in:

$$\frac{\frac{|k_z| a I'_m}{I_m}}{\omega^2 - \omega_{A0}^2} = \frac{\frac{|k_z| a K'_m}{K_m}}{\frac{\rho_e}{\rho_0} (\omega^2 - \omega_{Ae}^2)}$$

Then in order to make it identical to the one in the first exercise we write:

$$\frac{1}{\rho_0 (\omega^2 - \omega_{A0}^2)} \frac{|k_z| a I'_m}{I_m} = \frac{1}{\rho_e (\omega^2 - \omega_{Ae}^2)} \frac{|k_z| a K'_m}{K_m}$$

We have assumed the incompressible limit ($c_s \rightarrow \infty$) meaning that any thermal pressure perturbation propagates instantaneously. This assumption physically filters out pure acoustic waves, resulting in a system dominated by magnetic restoring forces. (Note: this simplifies both the differential equation and also the physics). For this reason the term with c_s is dominating over the other terms, at the end getting:

$$M_0^2 \rightarrow \frac{k_z^2 c_{s0}^2}{c_{s0}^2} = k_z^2 \quad M_e^2 \rightarrow \frac{k_z^2 c_{se}^2}{c_{se}^2} = k_z^2$$

Then since $\omega_A = k_z c_A$ we get:

$$\boxed{\frac{M_0 a}{\rho_0 (k_z^2 c_{A0}^2 - \omega^2)} \frac{I'_m(M_0 a)}{I_m(M_0 a)} = \frac{M_e a}{\rho_e (k_z^2 c_{Ae}^2 - \omega^2)} \frac{K'_m(M_e a)}{K_m(M_e a)}}$$